(5) Is phuzics "gradually becoming more scientific"?

The famous statistician Emmanual Parzen argues that Economics is becoming more scientific due to the trend that publication is becoming concentrated within a small group of researchers. We will test this hypothesis by investigating Hill estimator and Gini index.

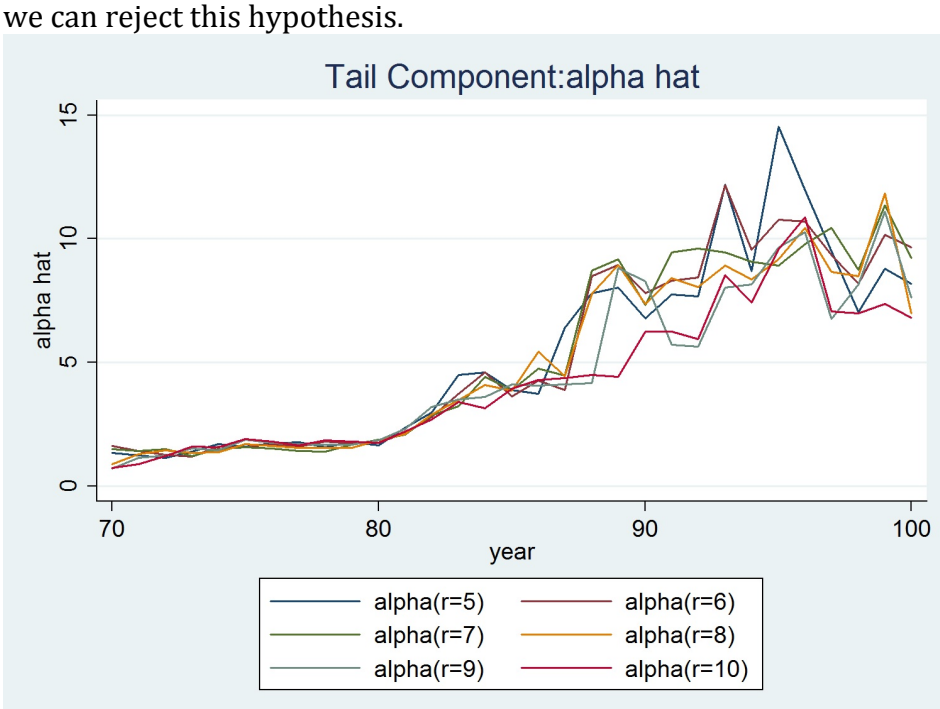
5.a Hill estimator.
Assume productivity of phuzicists follow Zipf's law:

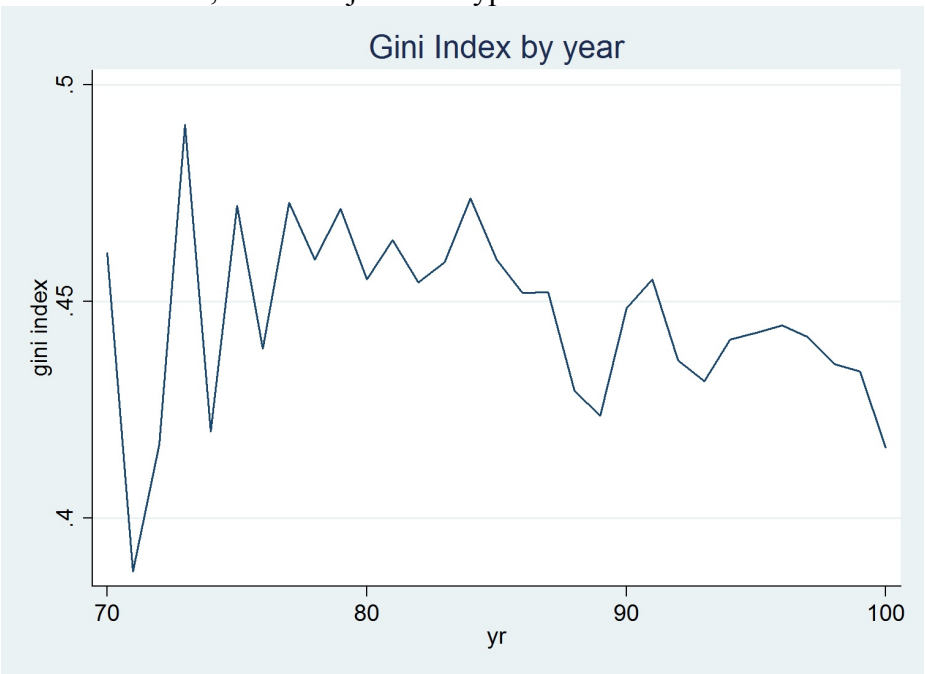
$$F(x) = 1 - (\xi/x)^{\alpha} \qquad x \geq \xi$$

The tail behavior is determined by tail component , which can be estimated by MLE(Hill's method):

$$\hat{\alpha}_r = (r^{-1}\sum_{i=1}^{r}\log(Y^{(i)}/Y^{(r+1)}))^{-1}$$

How to choose r is a problem. At year 70, there are only 11 samples, for some years, sample size is as large as 200. r shouldn't be too large, or some years' sample size would be too small; r shouldn't be too small, or it won't be able to catch the tail behavior; so we estimate α with six different r from 5 to 10.
The estimated α is plotted against years in fig "Tail Component: alpha hat". It's hard to justify which r should be used. However, all the estimated tail components show the similar trend. The estimated α is generally increasing by time with larger fluctuations. If Phuzics is becoming more scientific, estimated α should decrease, so we can reject this hypothesis.



5.b Gini coefficient:

We calculate the Gini index each year and plot in the fig "Gini index by year". If Phuzics is becoming more scientific, publications should increasingly concentrate in a small number of phuzicists, gini index should increase. However, the graph shows a downward trend, we can reject this hypothesis.